WORKED SOLUTION

STEP I 2018, Question 11

Mechanics • Pulley and rough inclined plane

STEP I 2018, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q11

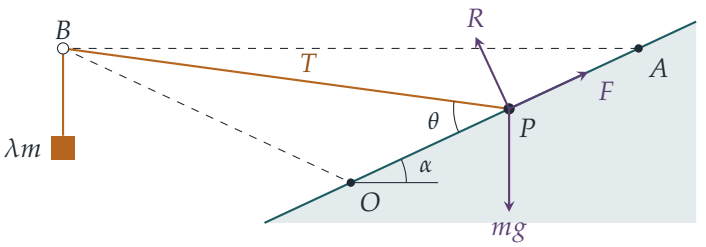
The key idea

Start with the geometry: $OA = OB = d$, so triangle OAB is isosceles, and because AB is horizontal its base angles are both α. The string runs from P back over O's side towards B, so it pulls P partly *down* the slope and partly *off* the plane. Friction must then hold the particle up, while the normal reaction is reduced by the tension. Each condition for equilibrium is an inequality in θ, and it has to hold for every θ in the range from (i), so we look for the worst position.

Solution

Model P and the hanging mass as particles, the string as light and inextensible, and the pulley as smooth, so the tension is the same throughout the string. The hanging particle is in equilibrium, so

$$T = \lambda m g.$$



Part (i)

Put O at the origin with the x-axis horizontal. Then $A = (d\cos\alpha,\ d\sin\alpha)$. The point B is at the same height as A and at distance d from O, so $B = (\pm d\cos\alpha,\ d\sin\alpha)$. The plus sign gives A itself, so

$$B = (-d\cos\alpha,\ d\sin\alpha).$$

This lies above the line of the plane, on the far side of O from A. In triangle OAB we have $OA = OB = d$, and $\angle OAB = \alpha$ because AB is horizontal and AO makes angle α with the horizontal. So $\angle OBA = \alpha$ too, and $\angle AOB = 180^\circ - 2\alpha$, which is obtuse because $\alpha < 45^\circ$.

Let P lie on OA. The string runs along PB, and the acute angle it makes with L is $\theta = \angle BPO$. This angle is acute because it is at most $2\alpha < 90^\circ$, as we now show. It is an exterior angle of triangle PAB, so

$$\theta = \angle BPO = \angle PAB + \angle PBA = \alpha + \angle PBA.$$

As P moves from O to A, the ray BP turns from BO towards BA, so $\angle PBA$ decreases from $\angle OBA = \alpha$ to 0. Therefore

$$\alpha \leq \theta \leq 2\alpha,$$

with $\theta = 2\alpha$ when P is at O and $\theta \to \alpha$ as P approaches A.

Note what this means for the forces. Measured from the down-slope direction PO, the string makes an acute angle θ and points away from the plane. So the tension has a component $T\cos\theta$ down the slope and a component $T\sin\theta$ perpendicular to the plane, pulling P away from it.

Part (ii)

Resolving perpendicular to the plane:

$$R = mg\cos\alpha - T\sin\theta = mg(\cos\alpha - \lambda\sin\theta).$$

Ask yourself Where is friction in this part, when we don't know μ?

We don't need it. Whatever μ is, P must stay in contact with the plane, so we need $R \geq 0$. (In fact $R > 0$: if $R = 0$ there would be no friction, and nothing would balance $mg\sin\alpha + T\cos\theta > 0$ along the slope.) So equilibrium at every point requires

$$\lambda\sin\theta \leq \cos\alpha \quad \text{for all } \theta \text{ that occur.}$$

The largest value of $\sin\theta$ occurs at $\theta = 2\alpha$ (or arbitrarily close to it, as P approaches O), since sin is increasing on $[\alpha, 2\alpha] \subset [0^\circ, 90^\circ)$. Hence $\lambda\sin 2\alpha \leq \cos\alpha$, that is $2\lambda\sin\alpha\cos\alpha \leq \cos\alpha$. Dividing by $\cos\alpha > 0$ gives

$$2\lambda\sin\alpha \leq 1,$$

as required.

Part (iii)

Gravity and the tension both pull P down the slope, so friction F acts up the slope. Resolving along the slope,

$$F = mg\sin\alpha + T\cos\theta = mg(\sin\alpha + \lambda\cos\theta).$$

Equilibrium at this position is possible if and only if $F \leq \mu R$ (this also forces $R > 0$, since $F > 0$). With $\mu = \tan\beta$, the condition is

$$\sin\alpha + \lambda\cos\theta \leq \frac{\sin\beta}{\cos\beta}(\cos\alpha - \lambda\sin\theta).$$

Multiplying by $\cos\beta > 0$ and collecting the λ terms:

$$\lambda(\cos\theta\cos\beta + \sin\theta\sin\beta) \leq \sin\beta\cos\alpha - \cos\beta\sin\alpha,$$
$$\lambda\cos(\beta - \theta) \leq \sin(\beta - \alpha).$$

Here $0 < \beta < 90^\circ$ and $0 < \theta \leq 2\alpha < 90^\circ$, so $-90^\circ < \beta - \theta < 90^\circ$ and $\cos(\beta - \theta) > 0$. So equilibrium at the position with angle θ is possible exactly when

$$\lambda \leq \frac{\sin(\beta - \alpha)}{\cos(\beta - \theta)}.$$

For this to hold at every position, it must hold where $\cos(\beta - \theta)$ is largest, that is, where θ is as close to β as the range $\alpha \leq \theta \leq 2\alpha$ allows.

Case $\beta \geq 2\alpha$. Every θ in the range is at most β, so $0 \leq \beta - 2\alpha \leq \beta - \theta \leq \beta - \alpha < 90^\circ$. Cosine decreases on $[0^\circ, 90^\circ)$, so $\cos(\beta - \theta)$ is largest at $\theta = 2\alpha$, the position at (or next to) O. A necessary condition is therefore

$$\lambda \leq \frac{\sin(\beta - \alpha)}{\cos(\beta - 2\alpha)}.$$

It is also sufficient, because at every other position the bound on λ is weaker.

Case $\alpha \leq \beta \leq 2\alpha$. Now $\theta = \beta$ lies in the range: by part (i), some position of P on L between O and A gives exactly this angle (approached as $P \to A$ if $\beta = \alpha$). There $\cos(\beta - \theta) = 1$, its largest possible value, and the condition becomes

$$\lambda \leq \sin(\beta - \alpha).$$

At that position the string is perpendicular to the total contact force, which makes angle β with the normal when friction is limiting. If $\beta = \alpha$, this forces $\lambda = 0$: the particle is then only just in limiting equilibrium under its own weight, and any string tension would make it slip.

Checks

- The two answers agree when $\beta = 2\alpha$: then $\cos(\beta - 2\alpha) = 1$ and both give $\lambda \leq \sin\alpha$.
- With $\lambda = 0$ the condition becomes $\sin(\beta - \alpha) \geq 0$, i.e. $\mu \geq \tan\alpha$, the familiar condition for a block to rest on a rough slope.
- For $\alpha = 20^\circ$, $\beta = 50^\circ$ the bound is $\sin 30^\circ / \cos 10^\circ \approx 0.508$. A computer check placed P at many points of OA, resolved the forces directly, and found that the largest λ giving equilibrium everywhere matched both formulae for thousands of random α and β. It also confirmed that $\alpha \leq \theta \leq 2\alpha$.

Common slips

- Putting B on A's side of O. Since $OB = OA$ and $B \neq A$, B is on the other side of O, so the tension pulls P *down* the slope, not up it.
- Forgetting that the tension reduces the normal reaction, which reduces the friction available.
- Checking only the end positions in the second case of (iii). The worst position is where $\theta = \beta$, which is in the interior.

Beyond STEP

The combination $\lambda\cos(\beta - \theta) \leq \sin(\beta - \alpha)$ is the algebraic form of the *angle of friction*: the total contact force must lie within angle β of the normal, the so-called cone of friction. Many friction problems collapse to a single line of geometry once you think in terms of that cone. Asking for equilibrium "for every position" is a small optimisation problem, and engineers reason the same way when they design for the worst case. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and the free-body diagram carefully and with many challenging problems.

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